Lisbon, Portugal

Lisbon's winter killers are cold homes and influenza, not El Niño. Vaccinate the housebound oldest at home before Christmas, and get the juntas' emergency fund paying again before the cold.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Lisboa district (2.28 million people)	16 in the district; about 4 in the city (our scaling)	Nov 2026 – Mar 2027 (heat figure: Sep – Feb)	Cold homes and influenza; flash floods

9/9 actions verified. 9 corrected, 0 unverified, 6 contraindicated (in *Do not do this*). Two proposals were dropped in verification: a weekly check-in program (no evidence that it cuts deaths) and a rainfall-triggered cash reserve (no such instrument exists in Lisbon). A donor top-up of the Fundo de Emergência Social was found contraindicated, because in 2023 the fund was underspent and the constraint is now a stalled protocol, not money. Exposure figure: Climate Impact Lab, September 2026, pre-peer-review, heat only. The city figure is our population scaling of the district figure, not an Impact Lab finding.

The premise. A historically strong El Niño is under way, but its measured effect on Lisbon's winter is weak: the city's rain follows the North Atlantic Oscillation, and Portugal's last two strong El Niño winters were among its warmest on record, with normal rain. So each Lisbon briefing runs as one winter season, from November 2026 to March 2027, against what kills people here whatever El Niño does: cold homes and influenza, which caused 4,685 excess deaths in Portugal last winter, and flash floods in the valley streets, which the new drainage tunnel will not be ready to stop this winter.

EXECUTIVE SUMMARY — AT A GLANCE



38.72°N 9.14°W, Portugal · © OpenStreetMap contributors (ODbL); UIT layer © Câmara Municipal de Lisboa; Natural Earth

- **The risk.** Cold and influenza caused 4,685 excess deaths in Portugal last winter. Valley-street floods kill in basements, and the drainage tunnel will not run this winter.
- **First action.** Lisbon's three health authorities (ULS) vaccinate housebound people aged 85+ at home by 15 December, from junta and Radar lists.
- **The money.** Floor €310,000, target €600,000, ceiling €910,000 (US$349,000 / 675,000 / 1.02 million). Home vaccination and night outreach buy the most.
- **Do not.** Do not heat closed rooms with braziers or gas heaters. Braziers were the leading cause of the carbon monoxide poisonings INEM reported in 2025.*

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* Suggested formulation: check with a health worker before sharing.

Resolution. This is one city-wide briefing for the Lisbon municipality (24 freguesias, 545,796 people, INE Census 2021). It is deliberately less granular than a set of zone briefings. Where the city differs inside, the actions name the freguesias.

WHAT LISBON SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort between now and the end of March. Backing lenses are shown per action. In Lisbon several lenses reached the same moves, but they leaned on the same few studies, so each card says where its evidence came from.

1. Vaccinate housebound people aged 85+ against influenza and COVID at home by 15 December, using the junta, Santa Casa, and Projeto Radar lists to find them

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OWNER	BY WHEN	COST	PREVENTS
ULS Santa Maria, ULS São José, and ULS Lisboa Ocidental	Lists by 15 Nov; doses by 15 Dec	About €150,000 (US$169,000), our estimate: 5,000 visits at €30. Vaccine already paid by the SNS	Influenza deaths in the age group that carried 60% of last winter's excess

Of Portugal's 4,685 excess winter deaths in 2025/26, 2,805 were people aged 85 or over (DGS seasonal report, via Lusa). Home vaccination is already allowed: DGS Norma 006/2026 lists the home as a vaccination site for the season running 30 September 2026 to 31 March 2027, and Orientação 004/2025 tells the ULS to use "every possible strategy" for bedridden people. The gap is delivery. People aged 85+ can be vaccinated only in SNS units, not in pharmacies, and coverage of people aged 65+ in the Lisbon metropolitan area was 75.7% in February 2026, against 80–85% in the Norte, Centro, and Alentejo (Vacinómetro). The lists exist: Projeto Radar (Santa Casa, CML, Social Security, PSP, and the juntas) holds about 30,000 records of older people living alone. Older people are most concentrated in Olivais, Ajuda, Benfica, Alvalade, and Alcântara (about 29–30% aged 65+ in 2011, the latest freguesia split found).

* Suggested formulation: check with a health worker before sharing.

Transferability: the mortality effect of influenza vaccination in old people is overstated by observational studies, because healthier people get vaccinated. DANFLU-2 (NEJM 2025) found high-dose no better than standard-dose (0.68% vs 0.73% hospitalized), so spend on reaching people, not on upgrading the product. No Portuguese cost per home visit was found; the €30 is our assumption, and the only published benchmark is a US median of US$56 per person (2013). The Radar protocol ends in 2026, so check it is renewed.

2. Sign the new emergency-fund protocol with every junta by 15 November, have health centers refer cold-home households to it, and write a named winter line into the 2027 budget before it goes to the Assembleia on 30 November

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ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CML, Social Rights (Direitos Sociais), with the 24 juntas	**Protocols 15 Nov; budget line by 30 Nov**	**€0 new money. The fund holds €4.19 million (US$4.71 million) to 2027, €2.93 million (US$3.30 million) of it in 2026**	**Households in cold homes going without heat, power, or food because the money is stuck**

The Fundo de Emergência Social (FES) is the only rail that already puts cash into Lisbon households: €1.37 million (US$1.54 million) to 6,286 households in 2023, about €218 each. CML amended it on 22 April 2026 and the Assembleia Municipal approved it on 19 May (Deliberação 249/AML/2026). The cap is €2,000 (US$2,250) per household a year, or €3,500 in exceptional cases. It pays electricity, gas, water, essential goods, and basic food. But the money is not reaching every junta. Alcântara's reports for the first and second quarters of 2026 say it has been waiting since October 2025 for the new protocol with CML, and it sends cases to Santa Casa meanwhile. Santa Maria Maior and São Vicente have been advancing their own cash. Sign every protocol before the cold, and use a budget alteration to raise the ceilings in the juntas where the need concentrates: Marvila, Santa Maria Maior, Arroios, and São Vicente.

* Suggested formulation: check with a health worker before sharing.

Transferability and limits: the FES is means-tested (household income per person at or below 70% of the minimum wage, with an emergent hardship). Age or illness alone does not qualify, so health-center referral works only for households that also pass the means test. The 2027 budget must reach the Assembleia by 30 November 2026 (Lei 73/2013 art. 45, as amended by Lei 66/2020); no proposal was public on 16 October. The mayor's coalition holds 8 of 17 seats, so a named line can be traded for votes.

3. Run a social-tariff sweep through the juntas before 30 November, moving eligible renters' electricity and gas contracts into their own names, and ask ERSE to set 1 November to 31 March as the winter period in which cut-offs of vulnerable customers are limited

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OWNER	BY WHEN	COST	PREVENTS
CML, Social Rights, with the juntas; ERSE for the rule	30 Nov 2026	Staff time only. The discount is paid by the energy sector	Heating rationed for cost, and winter disconnections

The social tariff takes 33.8% off electricity and 31.2% off natural gas, and it applies automatically, but only when the contract is in the eligible person's own name, for a permanent home, at 6.9 kVA or less. Eligibility is a listed benefit (CSI, RSI, social pension, and others) or annual income at or below €6,272.64 (US$7,059), raised for each member without income. Renters whose contract is in the landlord's name miss it. A sweep that brings a benefit certificate and the written lease to the supplier fixes that, and no junta or Lisboa E-Nova campaign doing this was found. On cut-offs: Decree-Law 130/2026 (29 June 2026) says interruptions of economically vulnerable customers in critical periods, "namely winter", are limited on terms ERSE is to set. It is in force, but no ERSE rule naming the months was found. The post-Storm Kristin disconnection ban (extended to 13 February 2027) covers calamity-area municipalities, and Lisbon is not among them.

* Suggested formulation: check with a health worker before sharing.

Limits: the tariff does not cover bottled gas, which many older Lisbon flats still use. Existing protections (notice before a cut, power reduction before disconnection, a payment plan after 60 days overdue) already apply; this action adds the months, not a new ban.

4. Register older and mobility-limited people living in basements and ground floors of the valley streets, door to door in November, each with a named neighbor who phones on an IPMA yellow rain warning and knocks on an orange one: "go up, not down"

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ADAPTATION INVENTORY: NO FLOOD-WARNING PRECEDENT WITH A MORTALITY OUTCOME

OWNER	BY WHEN	COST	PREVENTS
Serviço Municipal de Proteção Civil (SMPC), with the juntas of Alcântara, Santa Maria Maior, Misericórdia, Arroios, Campolide, and Benfica	30 Nov 2026	About €25,000 (US$28,000) in staff time, our estimate	Drowning below street level, the way flash floods kill here now

The one death of December 2022 was a woman in a flooded basement home in Algés, just across the city line in Oeiras. In Valencia in October 2024, 65 of the first 218 counted deaths were in homes and 32 in garages, and the court file shows most victims were lost between 19:00 and 20:00, before the 20:11 phone alert. Lisbon's flood footprint is the same valleys every time: the Alcântara valley (16 people rescued from homes and 6 from a supermarket on 13 December 2022), the Baixa, and the Almirante Reis axis. A list by name, and a neighbor who knows to act, is what reaches people before a warning does.

* Suggested formulation: check with a health worker before sharing.

Transferability: Valencia was a ravine flood and far larger, but the lethal setting, people below street level, is the same. Yellow warnings are frequent, so a knock on every yellow would wear out the neighbors; hence a phone call on yellow and a knock on orange. The register holds health data, so it needs a data-protection basis (GDPR) agreed with each junta. The €25,000 is our estimate.

5. Post night pump crews at Alcântara, the Baixa, Almirante Reis/Anjos, and Campolide no later than an IPMA orange rain warning, and on yellow when the warning covers the night; plan the season as if the drainage tunnel does not exist

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OWNER	BY WHEN	COST	PREVENTS
SMPC, with the Regimento de Sapadores Bombeiros (RSB)	Standing order by 1 Nov 2026	About €60,000 (US$68,000) of overtime for the season, our estimate	A repeat of 7 December 2022, when the red warning came after the rain

On 7 December 2022 Lisbon was under an orange warning from about 09:30. IPMA issued red at 22:26, and 47.8 mm fell in one hour at Tapada da Ajuda between 22:40 and 23:40, above the red threshold (over 40 mm in an hour). No SMS went to the public; IPMA's own president called the red warning "already reactive". On 12–13 December, 90.0 mm fell in six hours at Gago Coutinho, and the RSB logged 538 incidents. The tunnel built to drain the Baixa (Monsanto–Santa Apolónia, 4.4 km, designed for a 1-in-100-year rain) finished digging in July 2025. But its last section needs Santa Apolónia Metro station closed for about 415 days. On 7 October 2026 CML approved €2.82 million (US$3.17 million) to the Metro for that closure, still awaiting the Assembleia. The tunnel will not discharge this winter, and the Chelas–Beato tunnel is due in 2029.

* Suggested formulation: check with a health worker before sharing.

Thresholds (IPMA, Lisboa district): yellow 10–20 mm in 1 hour or 30–40 mm in 6 hours; orange 21–40 mm or 41–60 mm; red above 40 mm or 60 mm. Whether the municipal emergency plan (PMEPC) already pre-positions crews at a given level was not verified; if it does, this action is a check, not a change.

6. Keep night outreach to rough sleepers running every night from November to March, open shelter at a forecast minimum of 5°C or a storm warning rather than 3°C for 48 hours, replace the Santa Apolónia Metro shelter, take partners and pets, and vaccinate in shelters

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ADAPTATION INVENTORY: COLD-ALERT PROGRAM, TORONTO (NO EFFECT FOUND)

OWNER	BY WHEN	COST	PREVENTS
CML, Social Rights, with Comunidade Vida e Paz, Santa Casa, the ULS, and Metro de Lisboa	Trigger change by 1 Nov; rounds 1 Nov – 31 Mar	€160,000 (US$180,000) for Vida e Paz rounds: €1,060 a night × 151 nights	Cold exposure and missed flu shots among the 439 people sleeping rough

Lisbon counted 3,122 homeless people at 31 December 2024, 439 of them sleeping rough, concentrated around Avenida de Ceuta, Oriente, Santa Apolónia, and Anjos. The city's cold plan activates only on an IPMA yellow or orange cold warning with minimums of 3°C to −1°C for at least 48 hours. Lisbon's nights rarely get that cold, and the January 2026 activation that opened Santa Apolónia, Oriente, and Rossio stations at night came during Storm Ingrid, a storm, not a cold snap. Santa Apolónia is the station due to close for the tunnel works (action 5), so the plan needs a replacement site now. Comunidade Vida e Paz runs night rounds reaching about 503 people a night at €1,060 a day, served 179,550 suppers in 2025 for €419,902, says it receives no public funding, and reports fuel costs up 11% this year. Shelter vaccination has been done here before: 382 of 818 people in 2015/16. Partners and pets were accepted at the Lapa emergency shelter in February 2018.

* Suggested formulation: check with a health worker before sharing.

Evidence: no documented cold deaths among Lisbon rough sleepers were found, and the one evaluated cold-alert program (Toronto) found no effect on hospitalizations. The case rests on exposure, not on a measured effect. The grant keeps an existing service running through the season; the city's 2024–2030 homelessness plan (€70 million) funds year-round beds, not winter ones.

7. Fit sealed carbon monoxide alarms during Radar home visits to people aged 75+ who live alone, and combined smoke and CO alarms in overcrowded rented rooms in Mouraria and Martim Moniz, with cards in Bengali, Nepali, Hindi, Punjabi, Creole, and Portuguese handed out by the communities' own associations

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OWNER	BY WHEN	COST	PREVENTS
SMPC, with the juntas, Santa Casa (Radar), and migrant associations such as NIALP	Before 15 Dec 2026	About €50,000 (US$56,000): 2,000 alarms at about €25. Fitting not costed	Carbon monoxide poisoning and fire in the rooms people heat badly

INEM reported 28 carbon monoxide poisonings in 2025 up to 24 November, against 18 in all of 2024, with braziers (braseiras) the leading cause. Portugal's hospital admissions for CO poisoning peak from November to March. SMPC already runs a winter message banning braziers and grills for heating and giving the poison center (CIAV, 800 250 250) and 112. This action puts hardware and a visit behind the message. In Mouraria, migrant workers rent beds in subdivided rooms; the February 2023 fire on Rua do Terreirinho, in a flat where 16 to 22 people lived, killed two, which is why those rooms get smoke alarms too. Residents there trust their own associations and mosques more than the junta.*

* Suggested formulation: check with a health worker before sharing.

Evidence: there is no outcome study showing that distributing CO alarms reduces poisonings; the UK Be Gas Safe program (13,000 alarms) measured awareness only, and US follow-ups found batteries left unreplaced, hence sealed 10-year units. Sealed alarms retail at about €15–60. The brazier deaths found in the press were outside Lisbon.

8. Draught-proof and, in top-floor flats, insulate the homes of older people with a past heart or circulatory hospital stay, through a pooled fund held by the Cruz Vermelha Portuguesa, with Santa Casa asked to match it

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OWNER	BY WHEN	COST	PREVENTS
Cruz Vermelha Portuguesa (Lisbon), with health-center referrals	Start in Nov; first homes by Jan 2027	€240,000 (US$270,000): 150 homes at €1,600, the Vale Eficiência unit with VAT	Heart and circulatory deaths in cold homes, this winter and the next

Portugal had the highest excess winter mortality in the EU-14 (28%, Healy 2003), tied to poor housing thermal efficiency, and 15.6% of people cannot keep their home warm (Eurostat 2025, against 8.8% in the EU). The national Vale Eficiência voucher ended on 24 March 2026 with more than 28,000 eligible applications never awarded; its successor is not expected before 2027. In New Zealand, insulation cut all-cause mortality among people aged 65+ with a prior circulatory admission (hazard ratio 0.673, 95% CI 0.535–0.847; Preval 2017). Santa Casa had €108 million (US$122 million) in cash at the end of 2025, so it should be asked to match rather than given a grant.

* Suggested formulation: check with a health worker before sharing.

Transferability is weak, which is why this ranks eighth. The evidence comes from detached timber houses; Lisbon is mostly flats, where only top-floor homes can take ceiling insulation and renters need the landlord's consent. Heating added no benefit in the same study, and the respiratory subgroup result was not significant. Most homes will be done after the January peak.

9. Fund the Banco Alimentar Contra a Fome de Lisboa to buy staples in January, after its 28–29 November drive has filled the warehouses, and confirm the PESSOAS 2030 food card runs past January 2027 before moving anyone onto it

PARTIALLY CONFIRMED | FOOD | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CML, Social Rights, with the Banco Alimentar and the card's coordinating entities	Commit in Nov; buy in Jan 2027	Unfunded. No costed basis was found for an amount	Empty charity shelves in February, when heating bills peak

Lisbon will not run out of food this winter. The El Niño food signal here is weak: Portugal grows more rice and olive oil than it eats, and the September 2026 inflation spike was energy (up 15.3%), not food (unprocessed food up 3.7%). The break is household cash: the same euro goes to the gas bill. About 4.1% of residents were moderately or severely food insecure in 2022. The FES already pays for food, so no new guidance is needed (action 2). The PESSOAS 2030 card (€50.95 a month for the head of household) reached 49,110 people by April 2026, but it is valid only until January 2027. Watch INE's monthly unprocessed-food price index (act above 5% year on year) and DGEG's Monday diesel price.

Timing trap: buying in November duplicates the national drive on 28–29 November. Refood rescues surplus food with volunteers and is not a cash recipient. Who moves to the card is chosen by the coordinating entities, not by Social Security (ISS) alone.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	CML Social Rights	SMPC	RSB	Juntas	ULS (3)	Santa Casa	Vida e Paz	Cruz Vermelha	ERSE	Banco Alimentar	Assembleia Municipal
1. Home vaccination, 85+ housebound	**I**			**R**	**A/R**	**R**					
2. FES protocols and 2027 winter line	**A/R**			**R**	**C**	**I**					**C**
3. Social-tariff sweep and winter cut-off rule	**A**			**R**		**I**			**C**		
4. Basement register, "go up, not down"	**C**	**A**	**I**	**R**		**C**					
5. Night pump crews from orange; no tunnel		**A**	**R**	**I**							
6. Rough-sleeper season, 5°C trigger	**A**	**C**		**I**	**R**	**R**	**R**				
7. CO and smoke alarms, multilingual cards		**A**	**C**	**R**		**R**					
8. Draught-proofing and insulation				**C**	**C**	**C**		**A/R**			
9. January food purchase; card check	**A**			**C**						**R**	

What the chart shows: CML's Social Rights department is accountable for four of nine rows and the juntas do the work in seven, so the juntas are the delivery rail and their protocols (action 2) gate most of the rest. Civil protection owns three. Actions 2, 3, and 6 need a decision more than money: a signature, a sweep, and a lower trigger.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per euro. The envelope is bounded by what Lisbon can spend before the end of March, not by what it needs. Nothing here is committed; every placement is a gap.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
€310,000 (US$349,000)	**€600,000 (US$675,000)**	**€910,000 (US$1.02 million)**	**The cost of a home vaccination visit: no Portuguese figure exists**

Floor = home vaccination €150,000 + Vida e Paz rounds €160,000. Target = floor + draught-proofing €240,000 + CO alarms €50,000 = €600,000. Ceiling = vaccination at the lens's absorptive limit of €250,000 (about 8,300 visits at €30) + rounds €160,000 + 250 homes at €1,600 (€400,000) + 4,000 alarms (€100,000) = €910,000. The lens first proposed a target of €760,000; verification moved it to €600,000 (−21%). That is less than a third, but the change is structural. A €250,000 FES top-up was removed: in 2023 CML transferred €2.32 million and

* Suggested formulation: check with a health worker before sharing.

the juntas spent €1.37 million, so donor money would sit beside unspent public money. Draught-proofing rose from €200,000 to €240,000, because the €1,300 voucher unit is €1,600 with VAT.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
ULS Santa Maria, São José, and Lisboa Ocidental	Action 1: home vaccination visits	€150,000 (US$169,000)	Direct grant	5,000 visits × €30, our assumption; US benchmark US$56 per person (Community Guide, 2013). Vaccine paid by the SNS	Unverified cost
Comunidade Vida e Paz	Action 6: night rounds, 1 Nov – 31 Mar	€160,000 (US$180,000)	Direct grant	€1,060 a day × 151 nights (Vida e Paz, via RTP, July 2026). No public funding	Confirmed
Cruz Vermelha Portuguesa, Santa Casa to match	Action 8: draught-proofing and top-floor insulation	€240,000 (US$270,000)	Pooled fund	150 homes × €1,600 (Vale Eficiência €1,300 + VAT); NZ mortality evidence, weak transfer	Corrected
SMPC, through the juntas	Action 7: sealed CO alarms	€50,000 (US$56,000)	Pre-positioned stock	2,000 × €25 (Portuguese retail €18–58); fitting by Radar visits, not costed	Corrected

The funder's one thing: the cheapest life saved in Lisbon this winter is protected by a vaccine the state has already bought, sitting in a fridge. Pay for the last mile to the door. And do not put donor money into the emergency fund until CML has signed the protocols that let the juntas spend what it already holds.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here, or waste the window.

Do not heat closed rooms with braziers, charcoal, or portable gas or paraffin heaters, and do not hand them out *

CONTRAINDICATED

Braziers were the leading cause of the carbon monoxide poisonings INEM saw in 2025: 28 cases by 24 November, up from 18 in all of 2024. Portuguese CO admissions peak from November to March. People who cannot afford electricity heat this way.

Instead: safe electric oil-filled radiators paid through the FES (action 2), the social tariff to run them (action 3), and sealed CO alarms with a visit (action 7).

Do not count on the new drainage tunnel this winter

CONTRAINDICATED

The Monsanto–Santa Apolónia tunnel cannot discharge until Santa Apolónia Metro station has been closed for about 415 days. That closure had not started by mid-October 2026. The plan has slipped from 2019 to 2025 to "end of 2026" already. Lisbon has built drainage only after floods: the plan was voted weeks after the 2008 flood, and digging began a year after 2022.

Instead: run the December 2022 protocol unchanged (actions 4 and 5) until CML publishes a commissioning record.

Do not wait for the red warning, and do not tell people in basements to shelter in place *

CONTRAINDICATED

In December 2022 the red warning came at 22:26, after a 13-hour orange; the worst hour of rain began 14 minutes later, and no public SMS went out. In Valencia in 2024, most of the dead were lost before the 20:11 alert, many in ground floors and garages. Sandbags at a basement door can hold water in.

* Suggested formulation: check with a health worker before sharing.

Instead: act on orange, phone on yellow, and say "go up, not down; stay out of basements and garages" (action 4).

Do not run help for cold homes only when a cold-weather alert is issued

CONTRAINDICATED

Across 384 locations, moderate cold caused 7.29% of deaths and extreme cold 0.86% (Gasparrini 2015, Lancet), and deaths lag cold by one to three weeks. Toronto's cold-alert program found no effect on cardiovascular hospitalizations (RR 0.98). England's 75+ cold risk rose after its Cold Weather Plan. Lisbon's household cold deaths happen in ordinary chilly weeks, indoors.

Instead: a standing program from November to March (actions 1, 2, 3, and 8). Alert triggers stay right for rough sleepers, who face the cold directly (action 6).

Do not spread one-off heating vouchers across the city, or pour donor money into the emergency fund while its protocols are unsigned

CONTRAINDICATED

The Vale Eficiência ended in March 2026 with more than 28,000 eligible applications unanswered. NICE (NG6) found untargeted home-energy measures cost more than £100,000 per QALY on health grounds. In 2023 the FES spent €1.37 million of €2.32 million transferred, and in 2026 at least one junta still has no protocol to spend from. The FES cannot be released as an age-based tranche, because its regulation makes it means-tested.

Instead: sign the protocols and refer named households (action 2), and enroll them in the tariff that discounts every bill (action 3).

Do not brief or fund this winter as an "El Niño drought", or spend on mosquito control or heat now

CONTRAINDICATED

Portugal's last two strong El Niño winters, 2015/16 and 2023/24, were among the warmest on record with rain IPMA classed as normal, and Lisbon's four landmark floods (1967, 1983, 2008, and 2022) came in neutral or La Niña winters. *Aedes albopictus*, found in Lisbon in 2023, has no winter transmission. Climate Impact Lab projects about 4 extra heat deaths in the city for September to February.

* Suggested formulation: check with a health worker before sharing.

Instead: a winter-season briefing against cold, influenza, and floods. Revisit heat and mosquitoes in May, before summer.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Feb 1941	Cyclone, Lisbon and the Tagus shore (before the ENSO record)	"Several dozen" nationally (peer-reviewed); higher popular counts unverified	Not found
Nov 1967	Flash floods, Loures, Odivelas, Vila Franca de Xira; neutral	462 official, censored; more than 500 to about 700 estimated; about 4 in Lisbon city	The dead were in shacks in the flood channels; the shacks were cleared over decades
Nov 1983	Floods, Lisbon, Loures, Cascais; La Niña	About 10 dead; about 1,800 families homeless (press)	Not found
1998/99	Influenza and cold, Portugal; La Niña	8,514 excess deaths, nationally	DGS contingency plans for extreme temperatures from 2004
Feb 2008	Flash floods, Lisbon and Algés: 118.4 mm in a day, the record since 1864; La Niña	Lisbon deaths not verified	Drainage master plan (PGDL) voted weeks later, then shelved for lack of money
2014/15	Influenza and cold, Portugal; weak El Niño	5,591 excess deaths, nationally (76% flu)	Not found
2015/16	Very strong El Niño winter: third-warmest, rain classed normal	No event found	—
2016/17	Influenza, Portugal; neutral (ONI v6)	4,467 excess deaths, nationally	Not found

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Dec 2022	Flash floods, Alcântara, the Baixa, and Algés; La Niña	1 dead, in an Algés basement (Oeiras); €49 million damage in Lisbon; €47.4 million insured across the metro area	Tunnel digging began 2023; CML paid €87,000 (US$98,000) to 64 families by March 2023
2023/24	Strong El Niño winter: warmest since 1931/32, rain 92% of normal	No flood event found	—
Mar 2025	Storm Martinho, Lisbon and the Tagus Valley; neutral	No deaths found; 5,051 incidents in the region	Mostly wind and fallen trees
2025/26	Influenza and two cold spells, Portugal; neutral (ONI v6)	4,685 excess deaths to 8 April (+21%), 2,805 aged 85+, nationally	FES amended (April–May 2026); Metro stations opened at night in Storm Ingrid (Jan 2026)

ENSO phase from NOAA's Oceanic Niño Index (ONI v6), five overlapping seasons at ±0.5°C. Excess winter deaths are national; no Lisbon share was found. The recurrence nobody has noticed: none of Lisbon's deadly modern floods came in an El Niño winter, and every flood has drowned people in the city's worst housing, the 1967 shacks then and the Algés basement now. The city has built tunnels after each flood and never fixed those homes. Cold kills thousands nationally in the same housing every winter, in El Niño years and others alike.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Switch cold help on with an IPMA alert** (Emergency Manager, Economist) versus **run it all season** (Public Health, Skeptic). Resolved here by population: standing for people in cold homes, whose deaths come in ordinary weeks; triggered for rough sleepers, who face the night directly. A Lisbon evaluation of either would settle it, and none exists.
- **The basement register is a top action** (Emergency Manager, Historian) versus **flood money buys damage, not lives** (Philanthropist: the 2022 death was outside the city). The register stays because it is cheap and its lists also serve actions 1 and 7. It gets no donor money.
- **A warm El Niño winter lowers cold deaths** (the scope research) versus **influenza drives the toll whatever the temperature** (Public Health, Historian). The weak El Niño winter of 2014/15 was one of the worst flu winters; the warm 2023/24 winter still recorded excess deaths. Neither side can be tested before March.

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- **Weekly check-ins of the oldest living alone** (Frontline, Public Health) versus **no evidence that they cut deaths** (verification). Kept only as case-finding for actions 1 and 7, not as an action of its own.

WHAT NOBODY COVERED — BLIND SPOT

Hospital emergency capacity in the January influenza peak. The Emergency Manager named it ("the true body count is the flu wave in January") and no lens researched it. If emergency departments in the three Lisbon ULS cannot absorb the peak, every upstream action saves fewer lives than the cards suggest. Second: who lives below street level, and who heats with bottled gas, is not published by freguesia. The juntas and Santa Casa hold those lists; nobody else does.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, here the Lisboa district, of about 2.3 million people. They are not city or freguesia figures. The El Niño projection covers *heat only*; cold, influenza, and flood are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as the Lisbon municipality (Concelho de Lisboa: 24 freguesias, 545,796 people in the INE 2021 Census; INE's revised 2025 estimate of 658,236 has no freguesia split). This is one city-wide briefing, run at the commissioner's request to test whether it is sufficient before a set of five zone briefings; actions name freguesias where the city differs inside. The scope research of 16 October 2026 was reused, and what this briefing relies on was re-verified. Climate Impact Lab region, as the published map gives it: Lisboa (district). Mortality figures exist only at that scale.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline. The published map (Flourish 30298258, column *6mo_hotonly_total_mean*) gives Lisboa district 16 excess heat deaths for September 2026 to February 2027

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(by month: 1, 0, 3, 6, 3, 3) and Portugal 39. Scaled by population, 16 × 545,796 / 2,275,385 = 3.8, so about 4 for the city. That is our arithmetic, not an Impact Lab finding. It covers heat only and is never added to the Impact Map's long-run projections.

- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched Lisbon independently and in parallel, from web sources, to a fixed brief and the fixed premise above.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favor. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the end of March, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here this winter). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. ENSO labels were checked against the NOAA ONI v6 table.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted. Its 96 entries were read from the data embedded in the inventory page. None is for Portugal, and none covers influenza, vaccination, carbon monoxide, or flood warnings with a mortality outcome. The only cold-weather precedent, Toronto's cold-alert program, found no effect; it appears on action 6 and under *Do not do this*.
- **Money.** Costs are shown in euros with US dollars in parentheses, at €1 = US$1.1253 (European Central Bank euro foreign exchange reference rate, 16 October 2026).
- **Health wording.** Passages where a wording error could hurt someone (braziers, heaters, and carbon monoxide, and flood warnings and when to move to safety) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

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- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Estimates are labelled as ours. Every source is listed below.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in Lisbon was interviewed. No Lisbon-only excess winter mortality exists; every death toll for cold and influenza is national. The DGS 2025/26 winter figures come from press reports of a document not found online. Projeto Radar counts are from 2019–2022. Reservoir levels were not read (the APA server refused connections), and water is not a hazard this winter. The costs of home vaccination visits, the basement register, crew overtime, and alarm fitting are estimates. Whether the PMEPC already pre-positions crews at a given warning level, and whether every junta has now signed its FES protocol, were not verified.

References · status after verification in brackets

Premise and ENSO · NOAA CPC ONI v6 [confirmed] · IPMA winter 2023/24 bulletin [confirmed] · Pereira et al., EMS2026-669 (NAO and Portuguese rain) [confirmed] · Lisbon scope research, 16 Oct 2026

Cold and influenza · Renascença/Lusa on the DGS winter module 2025/26 [confirmed as reported] · DGS Norma 006/2026 [confirmed] · Observador, Vacinómetro coverage [confirmed, metro area] · Community Guide, home-visit vaccination · Healy 2003, JECH [confirmed] · Preval et al. 2017, BMJ Open [confirmed] · Chapman et al. 2009, JECH [confirmed] · Gasparrini et al. 2015, Lancet [confirmed] · Murage, Hajat, and Bone 2018 [confirmed] · Toronto cold-alert evaluation, Environ. Int. 2019 [confirmed] · Eurostat ilc mdes01 · Santa Casa, Projeto Radar [confirmed] · DECO, Vale Eficiência ended [confirmed]

Money and energy · Observador, FES amended [confirmed] · AML, Deliberação 249/AML/2026 [confirmed] · FES rules (FAQ) [confirmed] · Junta de Alcântara, Q2 2026 report [confirmed] · Correio da Manhã, FES 2022–23 [confirmed] · Lei 66/2020 [confirmed] · ERSE, social tariff (electricity) [confirmed] · ERSE, social tariff (gas) [confirmed] · Decree-Law 130/2026 [confirmed] · ERSE, Storm Kristin measures [confirmed] · Forbes Portugal, 2022 insured losses [confirmed] · Sábado, Santa Casa 2025 results [confirmed] · ECB reference rate [confirmed]

Floods · IPMA, 7–8 Dec 2022 [confirmed] · IPMA, 12–13 Dec 2022 [confirmed] · IPMA warning criteria [confirmed] · Observador, no SMS in 2022 [confirmed] · Correio da Manhã, 538 incidents [confirmed] · Postal, Algés death [confirmed] · Valencia DANA court file [confirmed] · Lisboa para Pessoas, tunnel excavation [confirmed] · DN, Chelas–Beato 2029 [confirmed] · ECO, Santa Apolónia closure [confirmed] · Fragoso et al., 18 Feb 2008 [confirmed] · RTP, 1967 floods [confirmed] · Sábado, flood chronology (1983) [press only]

Homelessness, CO, and food · RTP, 3,122 homeless [confirmed] · Observador, cold-plan trigger [confirmed] · Sábado, Metro night shelters [confirmed] · Comunidade Vida e Paz appeal [confirmed] · Lapa shelter, 2018 [confirmed] · RTP/INEM, CO cases 2025 [confirmed] · Acta Médica Portuguesa, CO admissions [confirmed] · DN, Mouraria fire [confirmed] · Banco Alimentar campaign [confirmed] · DECO, food card [confirmed] · ECO, INE September CPI [confirmed]

Exposure data · Climate Impact Lab, 451,000 at Risk · Sep–Feb heat-death map · Impact Map · Adaptation Inventory · Lisboa district population, Census 2021 (Lusa)

Locator: © OpenStreetMap contributors (ODbL); UIT layer © Câmara Municipal de Lisboa; Natural Earth.

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